

Notes: Lamont (JF, 1997)
Cash Flow and Investment: Evidence from Internal Capital Markets

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1 Big picture

- **Question.** Does an exogenous decline in a firm's cash flow or collateral value reduce investment in other parts of the same firm, even when those other businesses do not face worse investment opportunities?
- **Setting.** Lamont studies the 1986 collapse in oil prices, when crude oil prices fell sharply and oil companies experienced a large decline in cash flow and petroleum collateral value.
- **Key empirical object.** The paper tracks nonoil corporate segments owned by oil-dependent diversified firms. These nonoil segments operate in industries whose profitability is not positively related to oil prices.
- **Main identification idea.** The oil price shock hits the parent firm's oil segment, but the investment opportunity set of the nonoil segment should not directly deteriorate. If nonoil investment falls, the fall is evidence of within-firm financial interdependence.
- **Joint hypothesis tested.** First, the oil shock raises the cost of finance for the oil company by reducing cash flow/collateral. Second, internal capital markets transmit this financial shock from the oil segment to nonoil segments.
- **Main result.** Oil-dependent firms significantly reduce investment in nonoil segments after the oil shock, both in raw and industry-adjusted measures.
- **Subsidization result.** The investment cuts are concentrated in nonoil segments that were not self-financing in 1985, meaning segments whose investment exceeded their own cash flow before the shock.
- **Agency interpretation.** The evidence is consistent with diversified oil firms using cash-rich oil segments to subsidize underperforming nonoil businesses in 1985 and cutting those subsidies when oil cash flow collapsed in 1986.
- **Economic takeaway.** Internal capital markets transmit financial shocks across business units. Corporate segments are not financially independent stand-alone units inside diversified firms.

2 Incremental contribution of the paper

- **A natural experiment for internal finance.** The paper shifts the cash-flow-investment literature from cross-sectional comparisons of constrained and unconstrained firms to an identifiable shock: the 1986 oil price collapse.
- **Segment-level evidence.** Lamont uses COMPUSTAT segment data to observe investment inside conglomerates. This lets the paper examine how shocks to one part of a firm affect another part of the same firm.
- **Within-firm spillover test.** The paper does not ask whether oil firms reduce oil investment when oil prices fall. It asks whether oil-dependent parent firms cut investment in *nonoil* segments whose industries are not directly harmed by oil prices.
- **Joint test of external and internal capital markets.** The design requires both imperfect external finance and internal capital allocation. If either external finance were frictionless or segments were financially independent, nonoil investment should not respond.
- **Cash shortfall counterpart to cash windfalls.** Prior work such as Blanchard, Lopez-de-Silanes, and Shleifer studied cash windfalls. Lamont provides evidence from a large negative cash-flow/collateral shock.
- **Evidence on subsidization and overinvestment.** By showing larger cuts in previously subsidized nonoil segments, the paper connects financial constraints to the free-cash-flow/overinvestment literature on inefficient diversification.
- **Macroeconomic implication.** The paper identifies internal capital markets as a possible propagation mechanism: shocks to one sector can reduce investment in unrelated sectors through diversified firms.
- **Methodological contribution.** The paper demonstrates that corporate segment data can be used to study questions about financing constraints, capital allocation, and the theory of the firm.

3 Data and measurement

3.1 Oil-dependent parent firms

- The parent sample contains diversified firms with material exposure to oil and gas extraction before the 1986 oil shock.
- Ex ante oil dependence is measured using the share of 1985 cash flow and sales derived from oil extraction segments.
- The baseline sample contains 26 large oil-dependent firms that together own 40 nonoil segments.
- These are very large firms; the average 1985 sales in Table II is about 15 billion dollars.

Interpretation: The focus on large oil-dependent conglomerates makes the evidence especially striking because standard financing-constraints arguments are often thought to apply more strongly to small firms.

3.2 Nonoil segments

- Nonoil segments are corporate segments outside oil and gas extraction.
- The sample includes chemicals, railroads, forest products, agricultural seed, mining, metals, retail merchandising, transportation, steel, cement, and other industries.
- Industries are chosen so that segment profitability is not positively related to oil prices.
- Segment data report sales, capital expenditures, operating income, depreciation, and SIC codes.

Interpretation: The key test compares nonoil investment before and after an oil shock. The interpretation depends on nonoil investment opportunities not directly worsening when oil prices fall.

3.3 Investment and cash flow variables

- The main investment variable is capital expenditures divided by segment sales, denoted I/S .
- The primary change variable is $\Delta I/S$ from 1985 to 1986.
- Segment cash flow is operating income plus depreciation, scaled by sales.
- Industry-adjusted measures subtract the median value for comparable COMPUSTAT segments in the same industry.

Interpretation: Segment-level Tobin's q is not observed, so the paper relies heavily on industry adjustment to control for investment opportunities.

3.4 Control groups

- For most tests, the control group is the universe of comparable COMPUSTAT segments not owned by oil-extraction firms.
- Table VI also uses an alternative sample of firms that own oil segments but are not oil-dependent.
- The alternative sample addresses the concern that industries owned by oil companies have hidden oil exposure.

Interpretation: The control groups are designed to separate parent-level oil dependence from segment-level industry trends.

4 Models and empirical specifications

4.1 Standard cash-flow-investment equation

The paper begins from the standard investment-cash-flow framework:

$$I/K = a + bQ + cCASH/K + YearDummy + FirmDummy.$$

Interpretation: In firm-level data, the coefficient on cash flow may reflect both financing frictions and omitted investment opportunities. Lamont’s contribution is to use a shock to the parent firm’s oil cash flow that is plausibly unrelated to nonoil segment opportunities.

4.2 Segment-level change test

The main reduced-form comparison is

$$\Delta(I/S)_{segment, 1985-86} = (I/S)_{1986} - (I/S)_{1985},$$

with both raw and industry-adjusted versions.

Interpretation: A negative industry-adjusted change means that nonoil investment at oil-dependent firms falls relative to comparable nonoil industries.

4.3 Industry adjustment

For each nonoil segment,

$$\Delta(I/S)_j^{adj} = \Delta(I/S)_j - \text{median}\{\Delta(I/S) \text{ for matched industry segments}\}.$$

Interpretation: This adjustment is meant to remove shifts in investment opportunities and industry-level demand that affect all firms in the segment’s industry.

4.4 Ex ante and ex post shock tests

- Ex ante tests classify firms by 1985 exposure to oil cash flow and sales.
- Ex post tests classify firms by realized oil cash-flow declines and bond-rating downgrades.
- The paper compares these classifications to see whether cash-flow and collateral channels can be separated.

Interpretation: Bond downgrades are intended to proxy for a deterioration in collateral/borrowing capacity, while oil cash flow is a liquidity shock. The paper concludes that the two are hard to disentangle in this sample.

4.5 Subsidization regressions

The subsidy test estimates variants of

$$\Delta(I/S)_j^{adj} = \alpha + \beta Subsidy_j + \gamma \Delta(CF/S)_j^{adj} + \delta \Delta(S/S)_j^{adj} + \phi(I/S)_{1985,j}^{adj} + \varepsilon_j.$$

Interpretation: The coefficient on the subsidy dummy tests whether nonoil segments that previously relied on parent-level cash were cut back more sharply after the oil shock.

4.6 Oil-cash-flow and segment-cash-flow regressions

The paper also estimates level regressions separately in 1985 and 1986:

$$I/FS = \alpha + \beta(CF/FS) + \gamma(OILCF/FS) + \delta(S/FS) + \varepsilon.$$

Interpretation: In 1985, nonoil investment is related to oil cash flow; in 1986, it is related to own cash flow. This supports the story that oil cash flow subsidized nonoil segments before the shock but not after the shock.

5 Identification and empirical design

5.1 Core identification argument

- The 1986 oil price collapse was large and plausibly exogenous to individual oil firms.
- The treatment is parent-level oil dependence, not nonoil segment industry conditions.
- The outcome is nonoil segment investment, not oil segment investment.
- Industry adjustment removes typical changes in investment opportunities for the nonoil segment's industry.

Interpretation: The cleanest interpretation is a shock to internal finance. If nonoil segments are financially independent, their investment should not fall simply because another segment of the parent firm loses cash.

5.2 Threats to identification

- The sample is small: most key tables are based on 39 or 40 nonoil segments.
- Segment-level Tobin's q is unobservable, so industry adjustment is an imperfect proxy for investment opportunities.
- Oil cash flow and oil collateral value move together, making liquidity and collateral channels hard to separate.
- Tax changes in 1986 may affect the cost of capital and interact with oil-company profitability.
- Some nonoil segments may have hidden geographic or product-market exposure to oil.

Interpretation: Lamont acknowledges these limitations. The contribution is a focused natural experiment, not a large-sample structural estimate.

5.3 Why the evidence is still informative

- Nonoil cash flow generally rises in 1986, yet nonoil investment falls.
- Industry-adjusted investment also falls, suggesting the decline is not a nonoil-industry trend.
- A control group of diversified firms with oil segments but low oil dependence shows weaker declines.
- The largest cuts occur in segments previously subsidized by the parent.

Interpretation: The combination of these facts makes the internal-capital-market interpretation credible despite the small sample.

6 Section-by-section study guide

- **Introduction.** States the joint hypothesis: cash/collateral shocks reduce investment, and corporate segments are financially interdependent.
- **Section I.** Reviews the cash-flow-investment literature and explains why segment data and a natural experiment improve identification.
- **Section II.** Describes the 1986 oil shock and shows that oil extraction was hit much harder than nonoil businesses.
- **Section III.** Builds the oil-dependent parent sample and nonoil segment sample.
- **Section IV.** Shows that nonoil investment falls in 1986 in raw, industry-adjusted, and robustness specifications.
- **Section V.** Investigates why investment fell: collateral/cash flow, underinvestment versus overinvestment, subsidization, and own versus oil cash flow.
- **Section VI.** Suggests extensions, including plant-level data and other shocks.
- **Section VII.** Concludes that cash matters and segments are financially interdependent.

7 Tables and figures

7.1 Figures 1 and 2: Oil shock and asset-disposal environment

Read:

- Figure 1 shows real crude oil prices falling dramatically in 1986.
- Figure 2 shows nonenergy asset disposals by large energy companies.
- The two figures jointly establish that the oil shock was economically large and that energy companies were responding by changing portfolios and selling assets.

Detailed interpretation: The first figure is the paper's treatment plot: it demonstrates the timing and size of the external shock. The second figure helps interpret corporate responses: firms were not simply passively absorbing the shock; they were restructuring and disposing nonenergy assets.

Economic message: The oil collapse tightened financial conditions for oil firms and plausibly forced internal reallocation decisions.

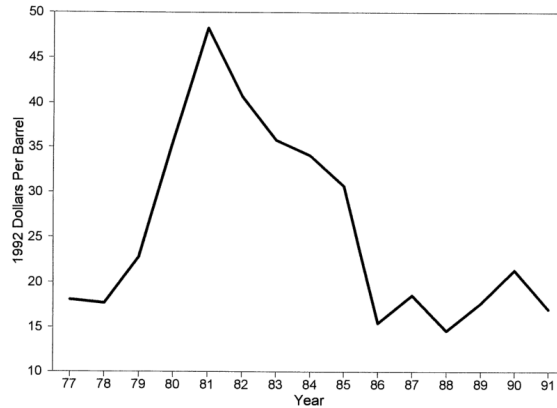


Figure 1. Real crude oil prices 1992 dollars per barrel.

(a) Figure 1. Real crude oil prices



(b) Figure 2. Nonenergy asset disposals

7.2 Tables I and II: Oil industry shock and oil-dependent firms

Read:

- Table I shows petroleum profit rates and operating statistics before and after the oil shock.
- Oil and gas production falls much more sharply than other petroleum-related activities.
- Table II lists the 26 oil-dependent parent firms and their ex ante oil cash-flow/sales dependence.
- The average firm is very large, with average 1985 sales over 15 billion dollars.

Detailed interpretation: Table I establishes that the shock is concentrated in extraction, while Table II establishes that the treatment firms are materially dependent on oil extraction. The treatment is therefore not vague “energy-sector exposure” but a measurable parent-level reliance on oil cash flow.

Economic message: Parent firms experience large liquidity/collateral shocks, creating the conditions for internal-capital-market spillovers.

Table I Profit Rates and Operating Statistics for Petroleum Companies, 1980–1990											
nies are a set of major petroleum companies that are tracked by the Department of Energy's Financial Reporting System.											
	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990
Panel A: Profit Rates for Lines of Business for FRS Petroleum Companies											
	15.3	12.4	7.7	7.4	6.9	5.5	3.0	3.6	7.2	6.4	6.8
m	17.5	16.1	12.7	10.3	9.4	9.4	3.0	4.9	6.3	5.8	7.9
as Production	20.9	20.2	14.0	11.3	10.8	9.5	0.8	4.1	2.8	2.9	8.5
nd Marketing	9.8	4.4	6.0	4.8	0.3	6.5	4.5	2.9	14.7	11.5	5.2
duction	15.1	15.6	20.8	16.6	20.8	15.0	13.2	12.8	9.6	10.2	11.2
as Production	23.0	17.7	11.8	14.1	13.3	13.8	12.8	9.5	9.9	8.7	12.5
nd Marketing	25.1	25.5	17.4	19.6	18.8	20.0	11.6	12.4	9.2	8.9	13.1
nal	26.4	9.0	4.7	7.7	4.5	3.3	16.3	4.7	11.6	8.0	11.2
ry	2.4	-1.1	-6.3	-13.2	-14.0	-19.0	5.3	-3.6	6.8	12.4	11.7
ther Energy	5.6	6.1	4.4	5.0	6.2	4.6	2.7	5.1	6.7	5.0	3.3
	-0.7	-6.8	-5.2	0.5	-1.8	-8.4	-0.8	0.5	-2.5	-2.3	1.9
	5.9	3.5	0.6	2.9	4.8	4.2	5.1	12.2	20.3	17.3	7.8
Panel B: Operating Statistics for FRS Firms and for US Petroleum Industry											
n, Crude Oil, And Natural Gas Liquids (Billion barrels)											
nies	2.09	2.07	2.08	2.06	2.09	2.12	2.09	2.07	2.10	1.91	1.81
ry	3.71	3.69	3.67	3.75	3.81	3.81	3.71	3.57	3.32	3.24	3.27
ut (Million barrels/day)											
nies	12.22	11.28	10.63	10.38	11.00	10.92	11.46	11.71	12.03	11.41	11.31
ry	15.35	14.66	14.01	13.69	14.27	14.19	14.93	15.09	15.43	15.65	15.91
npleted											
	926	1,159	625	513	597	539	258	356	224	363	347
it	1,917	2,081	1,580	1,586	1,882	1,776	647	911	822	922	1,164

(a) Table I. Petroleum operating statistics

Table II Firm Data: 1985–1986					
Ex ante cash flow and sales share are the percent of cash flow and firm sales that are derived from the petroleum extraction industry. The share of cash flow (sales) derived from the petroleum extraction industry is computed by dividing the total oil cash flow (sales) by the sum of the cash flow (sales) of all firm segments. Ex post change in net income and cash flow are the 1986–1990 dollar changes in firm net income and cash flow, divided by 1985 firm sales.					
Company	Ex ante		Ex post		Firm Sales (mil \$)
	1985 Percent of Cash Flow	1985 Percent of Sales	Δ Net Income	Δ Oil Cash Flow	
	Normalized By Firm Sales				
1 Amoco Corp	81	15	-4	-7	26,922
2 Atlantic Richfield Co	74	47	4	-1	21,723
3 Burlington Northern	28	28	-17	-2	8,651
4 Canadian Pacific	28	7	-2	-1	10,754
5 Chevron Corp	101	94	-2	-3	41,742
6 Dekalb Energy Co	77	28	-12	-17	570
7 Du Pont	49	12	1	-4	29,314
8 Fina Inc	71	8	-1	-5	2,403
9 Grace (W.R.) & Co	28	10	-12	-8	5,193
10 Homestake Mining	32	19	-0	-6	298
11 Imperial Oil Ltd	49	9	-4	-8	6,197
12 Kerr-McGee Corp	57	16	-13	-9	3,345
13 Litter Industries Inc	34	29	-5	-4	4,585
14 Mobil Corp	94	83	1	-3	55,960
15 Nova Corp of Alberta	41	29	6	NA	2,393
16 Occidental Petroleum	88	31	-4	-9	14,534
17 Phillips Petroleum Co	80	21	-1	-11	15,636
18 Placer Dome Inc	43	27	9	-7	302
19 Royal Dutch/Shell Grp	82	19	-0	-4	81,562
20 Schlumberger Ltd	94	65	-39	-17	6,119
21 Southdown Inc	49	19	-12	-5	325
22 Tenneco Inc	57	10	-1	-4	15,270
23 Union Pacific Corp	41	48	-12	-2	7,798
24 Unocal Corp	70	10	-1	-2	10,738
25 USX Corp-Consolidated	78	52	-12	-4	18,429
26 Zapata Corp	103	68	-62	-12	289
Average	63	31	-8	-6	15,040

(b) Table II. Firm data, 1985–1986

7.3 Table III: Segment Data for Oil-dependent Firms

Read:

- Table III lists the 40 nonoil segments owned by oil-dependent firms.
- The average change in investment-to-sales is -1.46 percentage points.
- The average change in segment cash-flow-to-sales is $+2.43$ percentage points.
- The segments span chemicals, railroads, forest products, agriculture, mining, metals, retail, steel, transportation, and other industries.

Detailed interpretation: This table is central because it shows the paradox motivating the paper: nonoil segment cash flow improves on average, but investment falls. That pattern is hard to reconcile with a simple investment-opportunity story and points toward parent-level financing constraints.

Economic message: Nonoil investment falls even though nonoil segment cash flow is not deteriorating.

Table III
Segment Data for Oil-dependent Firms

$\Delta I/S$ is the change in the segment investment to sales ratio between 1985 and 1986. $\Delta CF/S$ is the change in the segment cash flow to sales ratio between 1985 and 1986. Cash flow equals pretax operating profit plus depreciation. Expressed as percentage points. SIC is Standard Industrial Classification.

	Company	Segment	$\Delta I/S$	$\Delta CF/S$	1985 Size (Mil \$)	SIC Code
1	Amoco Corp	Chemicals	3.46	5.88	2905	2860 286
2	Atlantic Richfield	Spec & Int. chemicals	2.38	1.97	2155	2869 286
3	Burlington Northern	Forest products	-1.60	1.55	258	2411 241
4	Burlington Northern	Railroad	-6.63	-4.27	4098	4011 651
5	Canadian Pacific Ltd	Forest products	1.66	1.61	1546	2621 241
6	Canadian Pacific Ltd	Railroad	-3.40	-1.38	2408	4011
7	Chevron Corp	Chemicals	-1.30	6.05	2246	2869 286
8	Dekalb Energy Co	Agricultural seed	-2.85	-13.16	201	115 11
9	Du Pont	Ag-Ind. chemicals	-0.67	10.72	3388	2879 281
10	Du Pont	Biomedical products	0.19	3.08	1016	3844 384
11	Du Pont	Fibers	1.43	10.77	4483	2824 222
12	Du Pont	Indus.-cons. products	0.02	-0.65	2780	3861 367
13	Du Pont	Polymer products	-0.69	3.53	3379	2821 306
14	Fina Inc	Chemicals	-0.95	9.36	405	2821 286
15	Grace (W.R.) & Co	Specialty business	-0.91	0.42	787	2066 511
16	Grace (W.R.) & Co	Specialty chemicals	-1.21	-1.01	2254	2800 306
17	Homestake Mining	Gold	-16.64	12.11	169	1041
18	Imperial Oil Ltd	Chemicals	0.81	4.08	542	2860 287
19	Kerr-McGee Corp	Chemicals	-2.33	5.22	483	2812 281
20	Litton Industries	Adv. electronic	2.84	-5.65	1863	3812 367
21	Litton Industries	Marine engin. & prodtn	-0.32	0.05	975	3731 366
22	Mobil Corp	Chemical	-0.40	4.86	2266	3081 282
23	Mobil Corp	Retail merchandising	-0.88	2.57	6073	5311 596
24	Nova Corp of Alberta	Petrochemicals	6.92	2.09	541	2869 282
25	Occidental Petroleum	Agribusiness	0.40	0.37	6510	2011 651
26	Occidental Petroleum	Chemicals	-1.19	2.87	1621	2812 287
27	Phillips Petroleum	Chemicals	0.72	8.65	2266	2869 282
28	Placer Dome Inc	Mining	-0.43	1.10	221	1041 102
29	Royal Dutch/Shell Grp	Chemicals	-1.09	8.52	8583	2800 282
30	Schlumberger Ltd	Measurement & systems	0.51	0.13	1619	3820 737
31	Southdown Inc	Cement and concrete	-4.54	-0.29	265	3241 651
32	Tenneco Inc	Automotive parts	0.77	1.65	1074	3714 553
33	Tenneco Inc	Chemical	-1.87	2.34	841	2819 280
34	Tenneco Inc	Packaging	-0.72	0.25	851	2631 308
35	Tenneco Inc	Shipbuilding	-1.80	-0.00	1801	3731 361
36	Union Pacific Corp	Transportation	-4.39	6.87	3786	4011 421
37	Unocal Corp	Chemicals	-2.39	0.44	1217	2873 299
38	Unocal Corp	Metals	-9.41	-3.42	129	1099 106
39	USX Corp	Steel	-1.44	-8.72	6263	3312 101
40	Zapata Corp	Marine protein	-10.29	16.45	93	2048 207

Table III. Segment data for oil-dependent firms

7.4 Tables IV and V: Raw and industry-adjusted investment declines

Read:

- Table IV shows raw segment investment-to-sales falling from 7.94 in 1985 to 6.48 in 1986 and 6.24 in 1987.
- Table V shows a raw mean decline of -1.46 percentage points and an industry-adjusted mean decline of -1.41 percentage points.
- The median and number-positive tests also reject no decline.
- The decline is significant in both raw and industry-adjusted data.

Detailed interpretation: Table IV gives the raw investment fact. Table V shows the same fact survives industry adjustment, which is critical because the nonoil segments are in many different industries. The similarity between raw and adjusted changes implies that the fall is not merely a consequence of nonoil industry-wide investment declines.

Economic message: The central empirical result is a parent-finance-driven investment cut in unrelated nonoil segments.

Table IV
Raw segment I/S Level, 1985–1987

I is segment capital expenditure and *S* is segment sales. *I/S* expressed as percentage points.

	1985	1986	1987
No. of Observations	40	40	40
Mean	7.94	6.48	6.2
Median	6.91	6.13	5.8

the ex post magnitude of the cash flow decrease in the oil segment in 1986, and the ex ante likelihood that a large decrease would occur. In looking at ex ante

(a) Table IV. Raw segment *I/S* level

Table V
Change in I/S, 1985–1986

Dependent variable: $\Delta I/S$, where *I* is segment capital expenditure and *S* is segment sales. Expressed as percentage points. Median: The *Z*-statistic is the Wilcoxon signed-rank test, which tests the hypothesis that the observations are iid and symmetrically distributed around zero. Number positive: the 2-sided *p*-value is the probability of observing at most this number of positive or negative values, under the null hypothesis that the observations are independent and $\text{prob}[\text{positive}] = 0.5$. Industry-adjustment: For each observation of $\Delta I/S$, *I* subtract the median value of $\Delta I/S$ from a control group of COMPUSTAT segments that were in the same industry, but were owned by companies that did not have an oil extraction segment.

	Raw	Industry-Adjust
No. of Observations	40	39
Mean	-1.46	-1.41
<i>t</i> -statistic	(2.34)	(2.06)
<i>p</i> -value	(0.02)	(0.05)
Median	-0.90	-0.80
<i>Z</i> -statistic	(2.51)	(2.18)
<i>p</i> -value	(0.01)	(0.03)
Number positive	13	12
<i>p</i> -value	(0.04)	(0.02)

(b) Table V. Change in *I/S*, 1985-1986

7.5 Tables VI and VII: Robustness and cash-flow versus collateral evidence

Read:

- Table VI shows the investment decline persists under alternative normalizations and in an alternative control sample.
- Table VII asks whether investment cuts are explained by bond downgrades or ex post oil cash-flow shock severity.
- Downgrade and ex post shock variables have the predicted negative sign but are statistically weak.
- Adding 1985 investment explains a large share of variation.

Detailed interpretation: Table VI strengthens the investment-decline result. Table VII reveals a limitation: the paper cannot cleanly distinguish cash-flow effects from collateral/credit-rating effects. Both channels are plausible because the oil shock simultaneously reduces internal liquidity and petroleum collateral value.

Economic message: The investment decline is robust, but the proximate channel is bundled: liquidity and collateral both tighten.

Table VI

Alternative Measures of Industry-Adjusted ΔI , 1985–1986

All variables industry-adjusted and expressed as percentage points. *Alternative measures:* $(\Delta I)/S$ is the change in investment normalized by 1985 segment sales. $(\Delta I)/A_{85}$ is the change in investment normalized by 1985 segment total assets. $(\Delta I)/I_{85}$ is the percentage change in investment between 1985 and 1986. $\Delta(I-\delta)/S$ is the change in the net investment ratio (capital expenditures minus depreciation divided by sales) between 1985 and 1986. Median: The Z-statistic is the Wilcoxon signed-rank test, which tests the hypothesis that the observations are iid and symmetrically distributed around zero. Number positive: the 2-sided p -value is the probability of observing at most this number of positive or negative values, under the null hypothesis that the observations are independent and $\text{prob}[\text{positive}] = 0.5$. *Alternative sample:* Uses both the base sample of nonoil segments and an additional group of 36 nonoil segments. These 36 segments are owned by 17 firms who have an oil segment in 1985 but which are not oil-dependent in 1985 (less than percent of the firm's cash flow comes from oil). Mean: the coefficient on the dummy variable for oil-dependent firms in a regression including only a dummy and a constant. Constant term is shown. Median: the difference between the medians of the oil-dependent and not-oil-dependent sample. The Z-statistic is the Wilcoxon-rank sum test, which tests the hypothesis that the two samples have the same iid distribution. The reported 2-sided p -values are two sided.

	Alternative Measures				Alternative Sample
	$(\Delta I)/S_{85}$	$(\Delta I)/A_{85}$	$(\Delta I)/I_{85}$	$\Delta(I-\delta)/S$	$\Delta I/S$
No. of Obs.	39	39	39	39	75
Mean	-1.54	-1.09	-6.34	-1.74	-2.56
t -statistic	(2.31)	(2.26)	(0.48)	(2.47)	(2.49)
p -value	(0.02)	(0.02)	(0.63)	(0.02)	(0.01)
Median	-1.02	-1.32	-14.96	-1.25	-1.53
Z-statistic	(2.20)	(2.20)	(2.20)	(2.56)	(2.57)
p -value	(0.03)	(0.03)	(0.03)	(0.01)	(0.01)

(a) Table VI. Alternative measures of ΔI

Table VII

Ex Post Regression Evidence on $\Delta I/S$: Cash Flow vs. Bond Ratings

The dependent variable is industry-adjusted $\Delta I/S$, the change in the ratio of capital expenditures to sales for nonoil segments between 1985 and 1986. Downgrade is a dummy variable equal to 1 for the 17 segments whose parent firms had their debt downgraded by Moody's between 1985–86. Expostshock is a dummy variable equal to one for the 20 segments whose parent firms experienced a decline in oil cash flow between 1985 and 1986 that was greater than 4.3 percent of their 1985 sales. $\Delta CF/S$ is the industry-adjusted change in the cash flow to sales ratio between 1985 and 1986. $(\Delta S)/S$ is the industry-adjusted percent change in segment sales between 1985 and 1986. I_{85}/S_{85} is the industry-adjusted level of the ratio of capital expenditures to sales. $N = 39$. t -statistics in parentheses.

Constant	-0.99 (1.07)	-1.06 (1.06)	-0.46 (0.37)	-0.48 (0.37)	-1.16 (1.42)
Downgrade	-0.97 (0.70)		-1.13 (0.79)	-1.56 (0.99)	-0.86 (0.85)
Expostshock		-0.69 (0.50)	-0.89 (0.63)	-0.75 (0.47)	1.21 (1.17)
$\Delta CF/S$				0.00 (0.02)	-0.03 (0.30)
$(\Delta S)/S$				-0.04 (0.72)	0.01 (0.35)
I_{85}/S_{85}					-0.68 (7.24)

(b) Table VII. Cash flow versus bond ratings

7.6 Tables VIII and IX: Levels of investment and profitability

Read:

- Table VIII shows industry-adjusted nonoil investment is near industry norms in 1985 but below industry norms in 1986.
- Table IX shows nonoil profitability is below industry in 1985 and improves in 1986.
- Profit-to-sales rises significantly from 1985 to 1986.
- Cash-flow-to-sales also improves, though less strongly.

Detailed interpretation: These tables shift the interpretation from pure underinvestment to possible overinvestment/free-cash-flow behavior. In 1985, oil companies seem to invest normal amounts into below-average nonoil segments. In 1986, when oil cash falls, they cut investment and profitability improves.

Economic message: The evidence is consistent with a story in which cash-rich oil firms subsidized low-performing nonoil segments before the shock.

Table VIII
The Level of Investment

I/S is segment capital expenditure divided by segment sales, industry-adjusted, and expressed as percentage points. The last column is the difference in the first two columns. Median: The *Z*-statistic is the Wilcoxon signed-rank test, which tests the hypothesis that the observations are iid and symmetrically distributed around zero. Number positive: the 2-sided *p*-value is the probability of observing at most this number of positive or negative values, under the null hypothesis that the observations are independent and $\text{prob}[\text{positive}] = 0.5$. $N = 39$.

<i>I/S</i>	1985	1986	Δ 1985–1986
Mean	0.62	−1.05	−1.66
<i>t</i> -statistic	(0.75)	(1.95)	(2.50)
<i>p</i> -value	(0.46)	(0.06)	(0.02)
Median	−0.18	−1.49	−1.06
<i>Z</i> -statistic	(0.15)	(1.91)	(2.79)
<i>p</i> -value	(0.88)	(0.06)	(0.01)
Number positive	17	17	13
<i>p</i> -value	(0.52)	(0.52)	(0.05)

(a) Table VIII. Level of investment

Table IX
Profitability

π/S is segment operating income divided by segment sales. $CF/S = (\pi + \delta)/S$ is segment operating income plus segment depreciation divided by segment sales. Variables are industry-adjusted and expressed as percentage points. Median: The *Z*-statistic is the Wilcoxon signed-rank test, which tests the hypothesis that the observations are iid and symmetrically distributed around zero. Number positive: the 2-sided *p*-value is the probability of observing at most this number of positive or negative values, under the null hypothesis that the observations are independent and $\text{prob}[\text{positive}] = 0.5$. $N = 39$.

	1985	1986	Δ 1985–86
Panel A: π/S			
Mean	−1.82	0.29	2.11
<i>t</i> -statistic	(1.79)	(0.26)	(2.43)
<i>p</i> -value	(0.08)	(0.80)	(0.02)
Median	−2.58	0.82	2.34
<i>Z</i> -statistic	(1.69)	(0.53)	(2.75)
<i>p</i> -value	(0.09)	(0.60)	(0.01)
Number positive	16	22	28
<i>p</i> -value	(0.34)	(0.52)	(0.01)
Panel B: CF/S			
Mean	−2.05	−0.96	1.09
<i>t</i> -statistic	(2.00)	(0.88)	(1.38)
<i>p</i> -value	(0.05)	(0.39)	(0.17)
Median	−0.24	−0.20	0.48
<i>Z</i> -statistic	(1.63)	(0.61)	(1.31)
<i>p</i> -value	(0.10)	(0.54)	(0.19)

(b) Table IX. Profitability

7.7 Tables X and XI: Subsidies and cash-flow allocation

Read:

- Table X shows that previously subsidized nonoil segments cut investment much more after the oil shock.
- The subsidy coefficient is around −8 percentage points in the first two specifications and remains negative with controls.
- Table XI shows oil cash flow is related to nonoil investment in 1985 but not in 1986.
- In 1986, own nonoil cash flow becomes the robust predictor of nonoil investment.

Detailed interpretation: These tables provide the strongest internal-capital-market evidence. In 1985, nonoil investment is financed partly by oil cash flow. In 1986, when oil cash collapses, subsidies are cut and nonoil segments rely more on their own cash flow. This pattern is hard to explain without within-firm capital allocation.

Economic message: The corporate internal capital market reallocates cash across divisions in good times and withdraws subsidies in bad times.

Table X
Subsidies and $\Delta I/S$

The dependent variable is industry-adjusted change in investment to sales ratios for non-oil segments between 1985 and 1986. CF is segment cash flow $= \pi + \delta$. Subsidy is a dummy variable equal to one for the 5 segments which had $CF < I$ in 1985, where neither I nor CF have been industry-adjusted. $\Delta CF/S$ is the industry-adjusted change in the cash flow to sales ratio between 1985 and 1986. $(\Delta S)/S$ is the industry-adjusted percent change in segment sales between 1985 and 1986. I_{85}/S_{85} is the industry-adjusted level of the ratio of capital expenditures to sales. $N = 39$. *t*-statistics are in parenthesis.

	1985	1986	Δ 1985–1986
Constant	−0.37 (0.65)	−0.46 (0.76)	−0.58 (1.22)
Subsidy	−8.10 (5.07)	−8.32 (5.13)	−4.04 (2.64)
$\Delta CF/S$		−0.01 (0.12)	0.01 (0.12)
$(\Delta S)/S$		−0.04 (1.16)	0.01 (0.19)
I_{85}/S_{85}			−0.50 (4.91)

(a) Table X. Subsidies and $\Delta I/S$

Table XI
Levels of Investment and Segment and Oil Cash Flow, 1985 vs. 1986

FS is total firm sales. The dependent variable is $I/FS = I/S \cdot S/FS$ for nonoil segments, where I/S is industry-adjusted. S/FS measures the relative size of the segment by taking the ratio of segment sales to total firm sales. $CF/FS = CF/S \cdot S/FS$ where CF/S is the industry-adjusted nonoil cash flow to sales. $OILCF/FS$ is the (unadjusted) oil cash flow to firm sales ratio. $N = 39$. *t*-statistics are in parentheses.

	1985	1986	Δ 1985–1986
Constant	−1.21 (1.52)	−2.61 (3.24)	−1.11 (0.30)
CF/FS	0.35 (1.83)	−0.04 (0.19)	0.40 (3.55)
$OILCF/FS$	0.13 (2.40)	0.12 (2.62)	−0.00 (0.09)
S/FS		0.07 (3.46)	−0.00 (0.42)
Adj R^2	0.17	0.35	0.23

(b) Table XI. Investment and cash flow, 1985 versus 1986

8 Detailed result map

- **Oil shock.** Crude prices collapse in 1986, strongly reducing oil-company cash flow and collateral value.
- **Nonoil investment.** Investment in unrelated nonoil segments falls sharply in 1986.
- **Industry adjustment.** The decline survives adjustment for nonoil industry investment trends.
- **Robustness.** Alternative normalizations and alternative control groups support the result.
- **Mechanism.** The paper cannot fully separate cash-flow and collateral channels, but both imply tightened finance.
- **Subsidies.** Previously subsidized segments experience the largest cuts.
- **Cash-flow sensitivity.** Oil cash flow predicts nonoil investment in 1985, but own segment cash flow predicts it in 1986.
- **Interpretation.** The evidence supports financially interdependent internal capital markets and is consistent with free-cash-flow overinvestment before the shock.

9 Short summary

- Lamont uses the 1986 oil price collapse as a natural experiment for corporate cash-flow/collateral shocks.
- The treated units are nonoil segments owned by oil-dependent diversified firms.
- Nonoil segment investment falls significantly after the oil shock.
- Nonoil segment cash flow generally improves, weakening a direct investment-opportunity explanation.
- Industry-adjusted investment also falls, strengthening the financing interpretation.
- Robustness checks using alternative normalizations and control groups support the result.
- The decline is concentrated in segments previously subsidized by the parent.
- In 1985, oil cash flow predicts nonoil investment; in 1986, own segment cash flow predicts nonoil investment.
- The paper shows that internal capital markets transmit shocks across divisions.
- The evidence is consistent with cash-rich oil firms subsidizing underperforming nonoil segments before the shock.

10 Conclusion

10.1 Core conclusion

Lamont concludes that large decreases in cash flow and collateral value reduce investment. The evidence comes from nonoil segments of oil-dependent firms, which cut investment after the 1986 oil shock even though the shock was not a direct negative shock to their own nonoil industries.

10.2 Economic intuition

A diversified firm pools cash and collateral across segments. When the oil segment generates abundant cash, the parent can subsidize nonoil divisions. When the oil segment loses cash and collateral, the firm-wide shadow cost of funds rises, and nonoil divisions face tighter finance even if their own investment opportunities do not worsen.

10.3 Incremental contribution revisited

The paper’s incremental contribution is to use segment data and a commodity-price shock to test internal-capital-market spillovers. This is more direct than comparing cash-flow sensitivities across firms because the shock is specific to one part of the firm and the outcome is another part of the firm.

10.4 Implications for corporate finance

The results imply that corporate boundaries matter for real investment. Joining an oil business and a nonoil business in one corporation changes how financial shocks are transmitted. This has implications for diversification, capital budgeting, leverage, and the theory of the firm.

10.5 Implications for macroeconomics

Internal capital markets can propagate sector-specific shocks. A shock to oil extraction can reduce investment in chemicals, railroads, agriculture, steel, and other nonoil sectors through diversified corporate parents.

10.6 Limitations

The sample is small, segment-level q is unobservable, oil cash flow and collateral value move together, and tax changes may confound the cost-of-capital channel. The evidence is therefore best interpreted as focused, natural-experiment evidence rather than broad-sample proof.

10.7 Final takeaway

Cash matters, and corporate segments are financially interdependent. Internal capital markets can transmit shocks across unrelated divisions and can also support subsidies to weak segments when parent-level cash is abundant.